

Russell Dudek

MANUFACTURING STRATEGY & OPERATIONS EXECUTIVE

CONFIDENTIAL ORGANIZATION.
VICE PRESIDENT, GLOBAL MANUFACTURING STRATEGY

412.287.8640 | russelldudek@gmail.com | linkedin.com/in/russelldudek | <https://russelldudek.github.io/VPConfidential/>

EXECUTIVE PROFILE

Manufacturing strategy and operations executive with an uncommon combination of technical depth and operating scale. Advanced-electronics manufacturing, high-reliability quality, Amazon-scale customer-backward operations, robotics and machine-vision systems, materials-science grounding, and AI-enabled operating transformation. Builds the decision architecture, operating cadence, leadership routines, and cross-functional mechanisms that convert technical complexity into safer, more reliable, more scalable execution.

VERIFIED SCALE SIGNALS

13 years

Advanced electronics and high-reliability operations at Compunetics

~5 million

Second-day orders supported annually in Amazon Prime Pittsburgh operations

~1,000

Fielded robots informing current factory, customer, and machine-fleet feedback loops

100%

Safety record with no FAA incidents across regulated ZeusVu flight operations

ROLE-ALIGNED CAPABILITIES

Manufacturing strategy

Advanced technical operations

Safety / quality / delivery

Capacity and constraint visibility

Operating cadence

Continuous improvement

Digital operations

Cross-functional execution

Leadership systems

SELECTED EXPERIENCE

Vape-Jet

Director of Operations
2025-Present

- Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connects factory, field, customer, and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Operates in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.

Compunetics

Director of Operations and Engineering Management
2000-2013

- Progressed through R&D program management, production management, strategic acquisition, operations management, and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures, and process controls.
- Collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD; published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

Amazon

Operations Management / Site Leader
2014-2018

- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward operating environment supporting approximately five million second-day orders annually.
- Built leadership routines, labor planning, throughput visibility, escalation paths, standard work, and operating mechanisms across safety, quality, throughput, customer experience, and team engagement.
- Served on the Amazon Prime Now launch team with Whole Foods and contributed to logistics innovation through Prime Air research and Amazon Flex geolocation concepts.

Russell Dudek

MANUFACTURING STRATEGY & OPERATIONS EXECUTIVE

CONFIDENTIAL ORGANIZATION.
VICE PRESIDENT, GLOBAL MANUFACTURING STRATEGY

412.287.8640 | russelldudek@gmail.com | linkedin.com/in/russelldudek | <https://russelldudek.github.io/VPConfidential/>

MANUFACTURING LEADERSHIP THESIS

Standardize the operating contract. Differentiate the technical capability. Use one enterprise cadence for customer promise, capacity, capital, performance, escalation, and learning while protecting the process knowledge and local judgment that create technical advantage.

ADDITIONAL OPERATING EXPERIENCE

Cardinal Building Products Regional Director 2021-2023	- Built the company's first remote regional operation and modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows. - Advanced AI-enhanced transportation-management and route-planning concepts.
ZeusVu Chief Pilot, Autonomous Systems and Aerial Data Operations 2018-2022	- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents. - Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence. - Designed medical drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders, and governed execution.
DudeWorth Founder / AI and Automation Advisor 2017-Present	- Develops AI-readiness and opportunity frameworks that assess value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value. - Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails. - Advises on AI-enabled operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.

TECHNICAL AND OPERATING FOUNDATION

- Advanced electronics and process-driven manufacturing
- Materials science and physical chemistry grounding
- Safety, quality systems, root cause, standard work
- Lean, Gemba, Six Sigma, 5S and continuous improvement
- Capacity, labor, throughput and constraint visibility
- ERP/Odoo, Jira, Confluence, HubSpot, Slack
- Agentic AI, LLM workflows, RAG, human-in-the-loop
- Python, SQL, JavaScript, TensorFlow, Tableau, GIS

EDUCATION AND CREDENTIALS

University of Pittsburgh - Bachelor of Science
Academic grounding includes materials science, physical chemistry, and biology-related studies. Academic work is indexed through the Google Scholar profile listed below.

- Google AI Essentials
- Food Safety Management Certification
- IBM Enterprise Design Thinking Practitioner
- Six Sigma Certification
- OSHA 10

Google Scholar:
scholar.google.com/citations?user=yHQSVD8AAAAJ&hl=en

OPERATING MECHANISMS

- Customer promise and quality:** turn safety, delivery, service, standard work, root cause, escalation, and repeat-issue signals into owned action.
 - Capacity:** connect labor, assets, throughput, bottlenecks, and demand to operating choices.
- Technical and digital transition:** connect engineering, manufacturing, training, customers, ERP, telemetry, and AI-assisted work.
 - Leadership system:** explicit authority, cadence, coaching, accountability, capability development, and succession visibility.